S P E C T R A V I D E O
U S E R S   G R O U P -- Wgtn

Newsletter Number 19 -- March 1986

The next meeting is on Monday, March 17th at 7.00 in the Staff Common Room , Wellington Clinical School , Mein Street Newtown, the next will be on April 21st.

We regret that there will be no tutorial at that first meeting.

Wellington SV Users Group
Committee Members

Chairman ----- Darryl Alison Ph 85522 (Paraparaumu)

Secretary ----- Joseph Dol
Treasurer ----- Ross Fletcher 788454

Don Stanley 896379

Garry Clark 367872

Newsletter Committee
Steve Hanson 843495
John Matthews 327805

Other specific areas of responsibility for committee members are
- Don -- Software Consultant
- Garry -- Radio Access & public domain tape software;
- John -- Library;
- Darryl -- Program evaluation committee chairman & Kapiti users representative.

Club Subscriptions

The club subscription is now $15.00/year.

New members this month - Barry Warner and Mr E. N. Cowan. WELCOME!

Current Membership -- **161**.

Note that expiry date for subscriptions is printed on the address label of newsletter. Subscriptions not renewed a month after this date will not receive newsletters from then. Check your label !!

Newsletter Contributions

If you are submitting programs to go in the newsletter they MUST satisfy the following conditions or they will NOT be printed.

1 ... be typed/printed on A4 paper with about 5 lines left blank at top and bottom of page and an inch at least around the sides

2 ... typed with a dark ribbon OR printed using emphasized print mode

3 ... include a description of what the program does , typed on A4 paper with example input

4 ... In the case of programs above 20 lines in length they should be sent on cassette , in this case there is no need for the printout as I can merge them into the newsletter. Enclose a stamped addressed envelope for return of cassette

5 ... line numbers should be continuous -- do a RENUM before printing the programs

6 ... PLEASE MAKE SURE THEY WORK . and would readers please note that I am NOT responsible for programs sent by people which do not work .

7 ... In the case of screen dumps from 3d plotters please indicate the function used and try to get the print as dark as possible -- not always easy I know .

All other newsletter contributions should be sent to The Editor, P.O Box 26050, Newlands. They may miss the next newsletter if sent anywhere else. Articles should follow 1 and 2 above.

Deadline for Newsletter articles is the 30th/31st of the month previous to publication.

Please follow these guidelines, articles with a 4mm edge all around can not be printed , and neither can articles produced on a dot matrix printer with low resolution . They are simply unreadable after photocopying.

Palmerston North Subgroup

The Palmerston North subgroup are holding meetings on the 1st Monday of each month at THE COMPUTER EXPERIENCE in Palmerston North.

Club Sponsor

MicroStyle Computers -- 29 High Street, Lower Hutt
Action Computers -- P.O Box 7442, Wellington South.

Photocopying from Library

Anybody requiring articles from the library to be photocopied please send request to the LIBRARIAN, not any other committee member. We have had problems with articles not being copied when requested, please ensure that requests go to the librarian.

Also membership renewals should be sent to Box 26050, Newlands, Wellington, not to any other box, otherwise it is possible that the mailing list will not get updated.

SV 328 Owners:

A new book, The Magic of SpectraVideo is available now. $55.00 includes an assembler on tape. Very limited numbers available so be quick.

New Software available now - see advert elsewhere in newsletter.

And now another bit of news!

DOUBLE-SIDED BASIC FOR SV OWNERS

For those of you with access to double-sided drives, this is for you ...

For $20.00 you get:

- a disk
- FREE p&p
- ability to format double sided disks, under SV BASIC
- double sided BASIC
- full compatibility with single-sided BASIC
- 338K per disk (ss BASIC 168K)

AND, all your single-sided disks can be upgraded to double sided with no loss of data!!

This "ASTOUNDINGLY GENEROUS package" is a must !!!
Send name, address and $20.00 to BRIAN DOROFAEFF
42 MAHOE STREET, HAMILTON.

****

WANTED TO BUY

80 col-card, preferably around $140. Ph Hastings 798505
or write: P.D.Drew 83 Caernarvon Drive, Flaxmere, Hastings

NEWS FROM C.D.L.

We quote the following from Computer Distributors Limited Newsletter:

"SVI-318/328 Support.

Contrary to the rumours that are floating around CDL is NOT discontinuing SUPPORT on the 318/328 product range!! However we have shifted our priorities, due to market demand and future developments, to the MSX range. Market requirements on the SVI range do not warent the importation at present of all items as we are bound by substantial minimum order quantities. If sufficient orders do reach us on items that we do not have in stock at the time, we will put an order to SVI through. It's the same as what some dealers do: order when they have a sale only. Also our move to other premises is no indication with regards to our position as official SVI Distributors and Support to our products. Parties who believe our support leaves a lot to be desired are invited to come and see for themselves or talk to us. Constructive criticism is welcomed."

(In the light of this correspondence from CDL, we appologise to them for the comments made by our correspondent in our previous newsletter. It is clear that all efforts are being made under the prevailing business conditions to maintain support for SV users. The abovementioned comments were based on information which we considered reliable at the time of going to press. Once again our apologies CDL. ..EDITORS)

The move to other premises took effect on 1 March 1986 and they advise their new address is:

7 College Hill	Postal Address
Ponsonby, Auckland.	P.O. Box 31-355,
Phone (09) 397-499	Milford

Some software just released (Go see your dealer!!):

Lazy Jones

"A well known program for Commo-thing and Spectr(um) is now available for MSX. It's 39.95 and its about a cleaner in a Hotel who only wants to play games and does so in one of the 18 rooms in the hotel. He's haunted by the irate manager, thought of work and his cleaning trolley.. Excellent graphics and plenty of excitement."

The RS232I.OBJ program

"This program on the new X'Press system disk will initialise the RS-232 hardware in the 738. You need to execute this program first before the RS-232 commands are used. Type; BLOAD "RS232I.OBJ",r under MSX Disk Basic."

The Editfkey program

"It is advisable to execute the EDITFKEY.COM program on the X'Press MSX-DOS disk from the 80 column mode. So PLEASE type "WIDTH80" first in MSXDOS."

SAVING DISK SPACE IN BASIC

A Utility Program

by Graeme Dick

Disk Basic has 36 tracks available for File storage, with the capacity of each track being approximately 4K bytes. Regardless of how long your File is, (at least) one track is used. Many of the Basic programs available are quite short, and this means that much of the disk space is wasted.

One way of overcoming this to some extent is to merge a number of files and start off with a Menu so that you can call up the file you want. This enables you to fully utilise the disk space, and has even greater savings if back-up copies are kept, as they should be. There is a disadvantage in not seeing on the Directory exactly which programs are included in the file, but this can be overcome satisfactorily by combining similar programs, such as Disk Maintenance or Maths Games etc. Another disadvantage is that if the "Menu File" gets corrupted you may lose a number of programs, but then all you should have to do is go to the back-up copy!

The following program is an example of how to go about it. I have used the ON ... GOTO function in line 100 to go to the appropriate program. If you do this you need to insert an END at the end of each program or they will run on to the next one in the menu. You may prefer to use the ON ... GOSUB function, in which case each program should end with a RETURN and a line 110 should be inserted to CLS:GOTO 30. Program names need to be entered starting at line 21, the no. of programs in the menu in line 40, and the starting line number for each program in line 100. It's a good idea to put a REM statement before the start of each program to remind you what it is when looking at the listing. Use the line no. before the start of the program (e.g. line 199) so that the REM is never executed.

Programs to be merged must first be RENUMbered to start at the right line no. and then saved as ASCII files. Don't forget the END!

```
10 REM UTILITY PROGRAM
20 SCREEN0,0:  CLS:  IV$=CHR$(27)+"p":  NV$=CHR$(27)+"q":  I=0
21 DATA FAT
22 DATA cpm.zap
23 DATA FORMAT.LST
24 DATA SPINPATCH
25 DATA sysgen.bas
26 DATA format
30 LOCATE17,0:  PRINT"M E N U"
40 NP=6 'Enter no. of programs in Menu
50 FOR I=1 TO NP:  READ P$
60 LOCATE5,(2*I+3),0:  PRINT I;":   ";P$
70 NEXT I
80 LOCATE0,20:  PRINTIV$" Press  Program number  to continue "NV$;
90 LOCATE36,20:  INPUT P
100 ON P GOTO 200,1000,2000,3000,3500,4000 REM enter start line nos.
    here for each program
199 REM Program #1:  FAT (File Allocation Table)
200 PRINT"Enter Program #1 from here"
210 :
220 :
550 END:  REM Don't forget to insert an END at the END of each program
999 REM Program #2:  cpm.zap (Alters Function Keys for CPM)
```

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REPEATING A CP/M PROGRAM WITHOUT RELOADING IT FROM DISK

If you have just run a program, the A> prompt has reappeared, and you decide to rerun the same program, you type the program name and CP/M reloads it from disk before running it, even though the program is already in the memory. I often have the occasion to rerun programs such as STAT so I have all my CP/M disks set up so that I can rerun the last run program simply by typing REPEAT. The system doesn't reload the program.

All you need to do is to type SAVE 0 REPEAT.COM at the A> prompt. (Actually any other COM filename will do, eg. AGAIN.COM etc.) This puts an entry for REPEAT.COM in the directory but doesn't use any program storage space. Thereafter, whenever you type REPEAT (or AGAIN, or whatever you called your non-existant program) the system is fooled into rerunning the program already in memory.

Parameters can still be used. For instance, after STAT you can type REPEAT *.* and you will get the status of all the files on the disk.

This setup has the added advantage, to users of single-drive systems, that it is not necessary to have programs such as STAT.COM on every disk. To get the status of any disk which hasn't got this program on, do this:

1. Insert disk containing STAT.COM
2. Enter CTRL C, to log in the disk
3. Enter STAT and you load the STAT.COM program and get the status of the current disk.
4. Remove the disk and insert the disk whose status you want.
5. Enter CTRL C to log in the disk.
6. Enter REPEAT *.COM or whatever parameters you want to get the status of files on the second disk.

There was one occasion when, with a single disk system, this was the only way (apart from recopying the whole disk from the backup copy using 1COPY) to fix a disk on which one of the files had become corrupted. The file was read-only, the disk was full and so there was no room to put STAT.COM on the disk! Using the above procedure the corrupted file was set to read-write, then deleted, then replaced with a good copy.

Donald Rogers

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Alphabetical Index of WORDSTAR commands (v. 3.30)

by Graeme Dick

When first learning to use WS I compiled the following alphabetical index of the commands, partly to simplify learning but also to serve as a memory aid later on. Others have found it useful too, so here it is for your benefit. I make no claims about completeness (e.g. Mailmerge commands are only briefly referred to), but most are there. It's not meant to replace the manual, and you'll have to refer to this if you don't know how to use a command. My customised Print commands are listed (see previous article in Newsletter) - you may want to change these. If anyone wants a copy on disk, this can be made available at User Group meetings.

W O R D S T A R I N D E X

Abandon File		^KQ
Answering prompts:	delete character	^S or DEL
	restore character	^D
	delete entry	^Y
	restore entry	^R
	display File Directory	^F
	cancel a command	^U ESC
BLOCK MENU		^K
Boldface		^PB
Centre line		^OC
Change logged drive [CAUTION:WILL CRASH IF NO DRIVE]		L or ^KL
Change number of lines per page [No.=PL-(MT+MB)]		
Column block feature (toggle)		^KN
Compressed print ON		^PQ
Compressed print OFF		^PW
Copy block		^KC
Copy file		O or ^KO
Cursor: move	one character left	Backspace
	one character right	^D
	one line down (same column)	arrow (^X)
	one line up (same column)	arrow (^E)
	one word left	^A
	one word right	^F
	to bottom of screen (same column)	^QX
	end of block	^QK
	end of file	^QC
	last Block source or Find location	^QV
	left side of screen (column 1)	^QS
	place marker	^Qn (0-9)
	previous position before Save etc	^QP
	right side of screen	^QD
	top of block	^QB
	top of file	^QR
	top of screen (same column)	^QE
Delete:	Block	^KY
	character at cursor	^G
	character to left of cursor	DEL
	entire line	^Y
	File	Y or ^KJ
	line to left of cursor	^Q DEL
	line to right of cursor	^QY
	word right	^T
Display/Hide block		^KH
Display Print Control characters		^OD
Document File		D

Dot Commands: WORDSTAR

bottom margin	(Default=8)	.MBn
change footing margin	(2)	.FMn
change heading margin	(2)	.HMn
conditional page break		.CPn
create footing		.FO text
create heading		.HE text
left margin offset		.POn
non-printable line message		..text
omit page numbering		.OP
page break insert		.PA
paper length	(Default=66)	.PLn
placement of page number	(Def=col 33)	.PCn
resume page numbering		.PN
specify different page number		.PNn
top margin	(3)	.MTn

MAILMERGE

Ask operator for variable value	.AV
Display message	.DM
Insert document File	.FI
Print variable value in document	&name&
Read variables from Data File	.RV
Specify Data File for .RV	.DF

End paragraph		RETURN
Exit to system		X
Expanded print	ON (cancelled by carriage return)	^PE
File Directory (toggle)		F or ^KF
Find/Replace text:		^QA
	automatic replacement line by line	N
	backwards search	B
	carry out a specified no. of times	n
	global search	G
	ignore casing	U
	whole words only	W
	Repeat last Find/Replace command	^L
	Combined options possible: e.g. Global automatic replacement	GN
	* speed procedure by pressing another key to "freeze" the screen	
Find text:		^QF
	find specified occurrence	^QFn
Help		H
Help level Display/Set		^JH
HELP MENU		^J
Hide/Display block		^KH
Hyphen help	(toggle: Default=ON)	^OH
Insert feature (toggle: Default=ON)		^V
Insert Hard hyphen when soft-hyphen on		^P-
Insert one Return (not affected by line spacing)		^N
Italics (toggle)		^PY
Justify	(toggle: Default=ON)	^OJ

Left margin offset for printing (Default=8)		.POn
Line space set (1-9)		^OS
Locate next "Find" text		^L
MAILMERGE		M
Margins:	set Left (Default=column 1)	^OL
	set Right (Default=column 65)	^OR
	release margins (turns Word wrap OFF)	^OX
	set temporary margin at tab stop	^OG
Mark block beginning		^KB
Mark block end		^KK
Move block		^KV
Non-break space		^PO
Non-document File		N
ONSCREEN MENU		^O
Open Document File		D
Open Non-document File		N
Overprint character		^PH
Overprint line		^P RETURN
Page break display		^OP
Page break insert		.PA
Paragraph indent		^OG
Phantom rubout		^PG
Phantom space		^PF
Print:	File	P or ^KP
	stop/resume printing	P or ^KP
PRINT MENU		^P
PRINT FORMAT COMMANDS:		
	Boldface	^PB
	Compressed ON	^PQ
	Compressed OFF	^PW
	Double strike	^PD
	Expanded [cancelled by <CR>]	^PE
	Insert line feed without a <CR>	^PJ
	Italics	^PY
	Non-break space	^PO
	Overprint character	^PH
	Overprint line	^P RETURN
	Phantom rubout	^PG
	Phantom space	^PF
	Printer pause set in file	^PC
	Solid underline ON	^PA
	Solid underline OFF	^PN
	Strike out text	^PX
	Sub-script ON	^PV
	Super-script ON	^PT
	Sub-script OFF	^PV^PR
	Super-script OFF	^PT^PR
	Underscore [broken]	^PS
QUICK MENU		^Q

Read text into Open File from another File		^KR
Reform paragraph		^B
	continuously	^QQ^B
Rename a File		E or ^KE
Repeat a command:		^QQ^_
	vary speed (n=1-9, 1=fastest, def=3)	^QQ^_(n)
Review:	Dot commands and print control chars	^JD
	Flags in right-most screen columns	^JF
	Margins and Tabs	^JM
	Menu location of certain commands	^JI(i)
	Moving text procedure	^JV
	Paragraph reforming	^JB
	Place markers	^JP
	Ruler line	^JR
	Status line	^JS
Ruler display (toggle)		^OT
Ruler - set length from text		^OF
Run a programme		R
Save file and resume typing		^KS
Save file and return to NO-FILE MENU		^KD
Save file and exit Wordstar		^KX
Scroll:	up screen continuously by line	^QW
	down screen continuously by line	^QZ
	up one line (new text at top)	^W
	down one line (new text at bottom)	^Z
	up one screen	^R
	down one screen	^C
Set/Hide place markers		^Kn (0-9)
Soft hyphen	(toggle: Default=ON)	^OE
Solid underline ON		^PA
Solid underline OFF		^PN
Stop a command		^U
Strikeout		^PX
Sub-script ON		^PV
Super-script ON		^PT
Sub-script OFF		^PV^PR
Super-script OFF		^PT^PR
Tabulate:	cursor to tab setting	TAB
	clear tabs	^ON
	set tabs	^OI(i)
Underscore	[broken]	^PS
Vari-tabs	(toggle: Default=ON)	^OV
Write block to file		^KW
Wordwrap	(toggle: Default=ON)	^OW

THE WRATH OF GRAPES

by Graeme Dick

A recent addition to the public domain Basic programs held by the Wellington Users' Group is one called "Alcohol Absorption and Metabolism" in the catalogue. If you like puns, you too may prefer the title I gave the program (if you know your Ernest Hemingway then you'll understand).

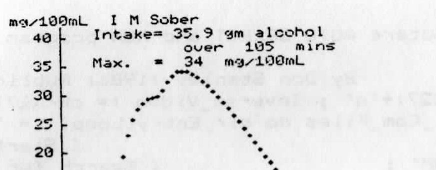
The program is meant to be educational, giving you an <u>indication</u> of your likely blood alcohol level after consuming alcohol. It takes a number of factors into account, apart from the amount consumed, such as body weight, sex (females will be affected to a greater extent than males for a given intake), and the time over which the alcohol is drunk.

I have included an example printout of the graph produced to show you what the program can do. The program uses the excellent SCREEN DUMP program written by Brian Parker of the Brisbane Spectravideo and MSX Users' Group to print the graph.

In this calculation (Mr) Sober (wt 70 kg) probably was! He had two cans of beer (good quality, export strength), and followed with a liqueur (40% alcohol) - a total of 35.9 gm alcohol over 1 3/4 hrs, which would have got him to about half the legal limit for driving. Of course, he might have been impaired even though under the limit, and probably would have been if an inexperienced drinker and/or driver.

It is important to remember that these sorts of calculations can never be better than approximations, but this program probably gives you a better idea than some other methods that have been promoted. I'm not going to say any more about it here - if you're interested, get a copy and run it. I'm sure you will be surprised in a number of ways.

I probably should add that the program has a good scientific basis, and gives a reasonable correlation with experimental results under similar conditions. For the last few years I have been involved in work on breath and blood alcohol analysis, and the program has followed from that involvement.



DRINK	%ALC	mL	Start	End
1	5.0	375	19.00	19.30
2	5.0	375	19.45	20.00
3	40.0	20	20.30	20.45

Metabolic rate = 15 mg/100 mL per hour

```
Program CPM_Autorun_Enhancement ;

{ Set this program up to autorun at the time a cold boot occurs in CPM. It
  will be run and will give the names of all COM files on the disk in drive
  A, plus a number for each. Select the number of the file you want to run,
  or 0 to just go to CPM. The selected program will be automatically run .

  This program is PUBLIC DOMAIN for CPM computer users. As it stands here it
  is configured for SpectraVideo CPM. The procedure Auto_Run is the only
  system dependent code, except in Initialize you may need to set the two
  variables NORMAL_VIDEO and INVERSE_VIDEO for your own hardware. They are
  set up here for a VT-52

  Don Stanley , Action Computers , February 1986
-------------------------------------------------------------------------}

Const Func1   = 17 ;        { BDOS Search First Function }
      Func2   = 18 ;        { BDOS Search Next Function  }
      Max_Com_Files = 100 ; { Maximum size of files array }

Type  Files_List = Array[1..Max_Com_Files] Of String[8] ;

Var   Wilds              : String[11] ;  { Filename to search for }
      N_Com_Files,                       { # of COM fils          }
      Loop,                              { Loop counter           }
      loop_4,                            {  "                     }
      Loop_2,                            {  "                     }
      Loop_3,                            {  "                     }
      FCB_Address,                       { File Control Block Addr}
      DMA                : Integer ;     { DMA Address            }
      Program_Number,                    { # selected By User     }
      Com_Files          : Byte    ;     { < 255 if more COM files}
      Dir_Entry          : Files_List ;  { Table of COM Files     }
      Normal_Video,
      Inverse_Video      : string[2] ;
{--------------------------------------------------------------------------

   Set up initial values for variables
--------------------------------------------------------------------------}

Procedure Initialize ;

Begin
  ClrScr ;
  GotoXY(10,10) ;
  Writeln('Action Computers AUTO SELECT and RUN program for CPM') ;
  Writeln;
  Writeln('             By Don Stanley (1986) Public Domain') ;
  Normal_Video := chr(27)+'q' ; Inverse_Video := chr(27)+'p' ;
  for Loop := 1 to Max_Com_Files do dir_Entry[Loop] := '        ' ;
  FCB_Address := $5C ;                          { Start of FCB }
  Wilds := '????????COM' ;               { Search for all COM files }
  For Loop := 1 to 11 do
    mem[FCB_Address+Loop] := Ord(Wilds[Loop]) ; { Load Filenames into FCB }
  mem[FCB_Address] := 1 ;                       { Select Drive A }
  DMA := $C000 ;                         { DMA Address for this program }
  Delay(3000) ;
End ;
```

```
{------------------------------------------------------------------------

     Set The DMA Address For CPM to read directory sector into

------------------------------------------------------------------------}

Procedure Set_DMA(DMA_Address : Integer) ;

Begin
  Bdos(26,DMA_Address) ;     { BDOS function 26 is set DMA address }
End ;

{------------------------------------------------------------------------

          Read the directory entries into Dir_Entry Array

------------------------------------------------------------------------}

Procedure Get_Directory_Entrys ;

Begin

  { ---------------  Find first COM file  ---------------}

  Com_Files := Bdos(Func1,FCB_Address) ;
  N_Com_Files := 1 ;

  { Now read the first file name into DIR_ENTRYS if COM files present }

  If Com_Files <> 255 Then Repeat
    Loop_4 := Com_Files*32 + DMA ;
    If mem[Loop_4] <> $E5 then Begin
      Loop_3 := 0 ;
      For Loop_2 := Loop_4+1 to Loop_4+8 Do Begin
        Loop_3 := Loop_3 + 1 ;
        Dir_Entry[N_Com_Files][Loop_3] := chr(mem[Loop_2]) ;
      End ;
    End ;

   { Now search for more COM files }

    N_Com_Files := N_Com_Files + 1 ;
    If Dir_Entry[N_Com_Files-1] = 'CPMBOOT ' Then
      N_Com_Files := N_Com_Files - 1 ;
    Com_Files := Bdos(Func2) ;
  Until (Com_Files = 255) ;
  N_Com_Files := N_Com_Files - 1 ;
End ;

{------------------------------------------------------------------------

   Give a list of files and let user select one of them to run

------------------------------------------------------------------------}

Procedure Put_Files_On_Screen ;

Begin

  { ----------------  Header  ---------------- }

  ClrScr ;
  GotoXY(22,1) ;
  Writeln(Inverse_Video,'   COM Files On Diskette   ',Normal_Video) ;
  Loop_3 := 1 ;
  GotoXY(1,3) ;
```

```
{ ----------------  File List  --------------- }

Repeat
  Repeat
    Write(' ',Inverse_Video,' ',Loop_3:3,' ',Normal_Video,
        Dir_Entry[Loop_3]) ;
    Loop_3 := Loop_3 + 1 ;
  Until (Loop_3 = N_Com_Files+1) or (Loop_3 mod 5 = 1) ;
  Writeln ;
Until(Loop_3 = N_Com_Files+1) ;
GotoXY(10,22) ;

{ ----------------  Get Users Number  --------------- }

Writeln(Inverse_Video,' Enter # Of Program To Run, 0 For CPM ',Normal_Video);
Program_Number := 0 ;
Repeat
  GotoXY(65,22) ;
  Read(Program_Number) ;
Until(Program_Number <= N_Com_Files) And (Program_Number >=0 ) ;
End ;

{-----------------------------------------------------------------------------

          Sort The Directory Entries From A -- Z

-----------------------------------------------------------------------------}

Procedure Quick_Sort_Directory( N_Com_Files : Integer;
                                Var k       : Files_List) ;

{ Sorts An Array (of TYPE FILE_LIST) into ascending order using a QUICKSORT.

  This procedure requires only the Var parameter changed to sort other types
  of array.

  The first parameter is the maximum element of the list to be sorted. You
  can sort a subset of the first X elements by setting this parameter to a
  value less than the maximum number of elements and by setting the array
  declaration below for LOWER & UPPER so that Max_Com_Files is replaced by
  your value  }

Label Step4,
      Step6 ;

Var Temp      : String[8] ;
    Top,
    i,
    j,
    lb,
    ub        : Integer   ;
    Lower,
    Upper     : Array[0..Max_Com_Files+1] Of Byte ;

Begin
    Top := 1 ; Lower[top] := 1 ; Upper[top] := N_Com_Files ;
    K[N_Com_Files + 1] := 'ZZZZZZZZ' ;
    repeat
      lb := Lower[top] ; ub := Upper[top] ; top := top - 1;
      While (UB > LB) do Begin
        i := lb ; j := ub ; temp := k[i] ;
```

```
Step4:  While (Temp < k[j]) do j := j - 1 ;
        If J <= i then Begin
          k[i] := temp ; Goto Step6 ;
        End;
        k[i] := k[j] ; i := i + 1 ; While (k[i] < Temp) Do i := i + 1 ;
        If j > i then Begin
           k[j] := k[i] ; j := j - 1 ; Goto Step4;
        End;
        k[j] := Temp ; i := j ;
Step6:  top := top + 1 ;
        if i-Lb < ub-i Then Begin
           Lower[Top] := i + 1 ; Upper[top] := Ub ; ub := i - 1 ;
        End Else Begin
           Lower[top] := lb ; Upper[top] := i - 1 ; lb := i + 1 ;
        End ;
      End ;
    Until (Top = 0 ) ;
End ;

{-----------------------------------------------------------------------

    Remove any file names which occur more than once

-----------------------------------------------------------------------}

Procedure Chuck_Out_Duplicate_File_Names ;

Begin
  Loop_2 := 1 ;
  Repeat
    If Dir_Entry[Loop_2] = Dir_Entry[Loop_2+1] Then Begin
      For Loop_3 := Loop_2 to N_Com_Files-1 Do
          Dir_Entry[Loop_3] := Dir_Entry[Loop_3+1];
      N_Com_Files := N_Com_Files - 1 ;
    End ;
    Loop_2 := Loop_2 + 1 ;
  Until (Loop_2 >= N_Com_Files) ;
End ;

{-----------------------------------------------------------------------

         Auto Run the selected COM file

-----------------------------------------------------------------------}

Procedure Auto_Run ;

Begin

  { What happens in here is that we place the file name which the user has
    selected, plus a RETURN, into the keyboard buffer. Then do a warm boot
    and the system interprets the characters in the buffer as the program to
    run !!!

    This code is specific to SpectraVideo CPM computers. To convert to other
    CPM hardware you need to know

      ( 1 ) Keyboard buffer pointer to start -- FA1A/1B here
      ( 2 ) Keyboard buffer pointer to end   -- FA1C/1D here
      ( 3 ) Keyboard buffer start            -- FD8B    here
      ( 4 ) Warm Boot Address                -- 0000    here    }
```

```
  Clrscr ;
  Loop_2 := $fd8B ;
  For Loop := 1 to 8 do Begin
    mem[Loop_2] := Ord(Dir_Entry[Program_Number][Loop]) ;
    Loop_2 := Loop_2 + 1 ;
  End ;
  mem[Loop_2] := $0a ;
  Inline($f3/            { DI          }
         $3e/$94/        { LD    A ,$94   }
         $21/$1a/$fa/    { LD    HL,$FA1A }
         $77/            { LD    (HL),A   }
         $23/            { INC   HL       }
         $3e/$fd/        { LD    A,$FD    }
         $77/            { LD    (HL),A   }
         $23/            { INC   HL       }
         $3e/$8B/        { LD    A,$8B    }
         $77/            { LD    (HL),A   }
         $23/            { INC   HL       }
         $3e/$fd/        { LD    A,$FD    }
         $77/            { LD    (HL),A   }
         $c3/00/00) ;    { JP    0000     }
End ;

{---------------------------------------------------------------------

             We Have No COM Files So Tell And Stop

---------------------------------------------------------------------}

Procedure Tell_User_No_Com_Files_Exist ;

Begin
  Clrscr ;
  GotoXY(10,10) ;
  Writeln(Inverse_Video,' No COM Files Exist On The Diskette In Drive A',
          Normal_Video) ;
End ;

{ -----------------------  Runs From Here  ----------------------------}

Begin
  Initialize ;                                { Set up variables }
  Set_DMA(DMA) ;                              { Set up DMA Address }
  Get_Directory_Entrys ;                      { Get a List of files from A }
  If N_Com_Files > 0 Then Begin
    Quick_Sort_Directory(N_Com_Files,Dir_Entry) ;  { Sort the list of files }
    Chuck_Out_Duplicate_File_Names ;          { Guess what this does };
    Put_Files_On_Screen ;                     { Print them on screen }
    If Program_Number <> 0 then Auto_Run ;    { Run selected program if non-0 }
    Set_DMA($80) ;                            { Reset DMA and back to CPM }
  End
  Else Tell_User_No_Com_Files_Exist ;         { Hey, No Files on here  }
End.
```

===

Oh how rewarding for us .. someone has admitted that our modest newsletter is providing INSPIRATION !!!
The following article from Deane Whitmore was inspired by our WORDSTAR patches article in our previous issue. Based on a magazine article, these further patches increase its speed and perform other tasks not available from the installation program supplied with the package.

(See Over =>)

CUSTOMIZING "WORDSTAR" - Version 3.30

To customize Wordstar, you have to patch it - change individual bytes in the main Wordstar program, WS.COM, and thereby change the instructions that Wordstar gives to the hardware inside your Spectravideo. When you are ready to begin patching, you can use either the Wordstar installation program or DDT.COM. Each has its advantages: With the installation program, you gain the use of mnemonic labels; with DDT you gain speed and access to extra patching areas unavailable from the installation program. If you patch Wordstar using DDT.COM and the CP/M's built-in SAVE command, you gain two advantages. Once you're comfortable with DDT, you can patch and repatch Wordstar very quickly. You can also patch very large routines onto the end of WS.COM, making it larger than the original. With DDT, you can patch anything and create some spectacular results.

An example use of DDT used to patch Wordstar follows:

Load DDT and WS.COM.

* At the CP/M prompt, enter **DDT WS.COM <ENT>**
* DDT will load itself and then load WS.COM. When it's ready, it will sign on and display the following information:

NEXT PC
4600 0100

Patch a new value into WS.COM

* Suppose you want Wordstar to clear the screen when you exit and return to CP/M. To do that you need to insert two bytes at the "TRMUNI" (TeRMinal UNInitialization) label at address 027Eh.
* So, at the DDT hyphen (-) prompt, type **S027E <ENT>**
  DDT replies with: 027E 00.
* To change it, type in a new value. In this case, enter the value 01h and hit ENTER: 027E 00 **01 <ENT>**
* DDT will move to the next address and show the value stored there: 027F 00.
* This time, enter the value 1Ah: 027F 00 **1A <ENT>**
* DDT will move to the next address (0280h), but you're finished with this particular patch. To get back to DDT's hyphen (-), just enter a period (.): 0280 00 **. <ENT>**

From this point you can go on to patch other areas in WS.COM by using the "S" command in the same way, as many times as you need to. You can even go back and patch the same areas over again; nothing is permanent until you take the next step.

Exit DDT and save the patched WS.COM

* Exit DDT by typing ^C (control-C) at DDT's hyphen (-) prompt; the computer will warm boot, and you'll be back at CP/M's A> prompt.

Now before you do anything else

* You must use CP/M's built-in SAVE command to save your patched WS.COM from memory into a disk file. I suggest you save it under a new name, such as WSX.COM. If you save it as the name WS.COM, you'll wipe out your unpatched WS.COM, which you may need again if your patches don't work. So save it as WSX.COM and test it. If it works then rename it to WS.COM.
* If you haven't made WS.COM any longer than it was by tacking a patch on the end then type: A>**SAVE 69 WSX.COM.**
* If you now load WSX.COM and then exit it, the screen will clear everytime before displaying the CP/M A> prompt.

MAJOR PATCH POINTS IN WORDSTAR

DELCUS - Address (028Eh) - Sets the length for a delay after the cursor is moved to a new spot. Lower this value to shorten the delay. My version of Wordstar has the value of 01h.

DELMIS - Address (028Fh) - Sets the length for a delay after miscellaneous screen functions. Lower this value to shorten the delay. if you lose characters or find odd things happening on the screen, it's too short.

DEL1, DEL2 - Address (02AFh,02B0h) - Sets length for short delays. These two values set the cursor blink rate when the cursor is on a highlighted character. they also control how the cursor jumps back and forth between the ^QA "REPLACE Y/N:" prompt and the text to be replaced.

DEL3 - Address (02B1h) - Sets length for a medium delay. This value controls how long Wordstar waits after you press a prefix key (^K,^Q,^P,^O,^J) until it displays the menu.

DEL4 - Address (02B2h) - Sets the length for long delays. This value controls the time for the "abandon file" and "new file" messages and for the Wordstar's sign-on messages, including the MicroPro copyright message.

DEL5 - Address (02B3h) - Sets the delay for screen update after a keystroke and (with DEL 4) affects screen update after a horizontal scroll.

SCRLSZ - Address (02BAh) - Sets the number of columns Wordstar will move during a horizontal scroll. the standard value here is 14h, which means you scroll 20 columns. You can get faster horizontal scrolling if you increase this value to 28h for a 40 column scroll and shorten the delays at DEL4 and DEL5.

ITHELP - Address (034Dh) - Sets the default help level at boot up. In a standard installation this byte is set to 03h for help level 3. Patch to 02h,01h,or 00h for help levels 2,1, or 0, respectively.

INITWF - Address (0371h) - Turns hyphen-help on or off. The standard setting is FFh, which turns hyphen-help on. Patch this byte to 00h to turn hyphen-help off.

ITPOPN - Address (03FDh) - Turns page-numbering on or off. The standard setting is 00h, which causes page numbers to be printed, unless you specify otherwise with a dot command. Patch this byte to FFh if you want the default to be no page numbers.

PATCHES FOR DOT-MATRIX PRINTERS

The following patches are for any Epson compatible dot-matrix printer. All the standard settings made by Wordstar can be patched, but the most frequently changed are the user defined codes (^PQ,^PW,^PE,^PR). There are five (5) bytes to each area, the first byte is the number of bytes that follow,a maximum of four (4) to each area. do not over-run these four bytes as garbage will be sent to the printer.

USR1 (^PQ) - Address (06CFh) - Lets use double width printing. Double width printing requires one string to turn it on and another string to turn it off. To turn it on, patch USR1 with these bytes: **03h 1Bh 57h 01h.**

USR2 (^PW) - Address (06D4h) - To turn double print off, patch USR2 with these bytes: **03h 1Bh 57h 00h.** Now you can enter ^PW to turn off double-wide printing and revert to your previous print pitch.

USR3 (^PE) - Address (06D9h)

USR4 (^PR) - Address (06DEh)

The above two addresses can be patched with whatever printer function you desire. Remember not to exceed the five bytes per location.

Next month I will give you further patches to enhanced a dot matrix printer and improve the screen display of Wordstar at boot up.

PUBLIC DOMAIN DISKS

Our previous few issues contained lists of new software, now available on Public Domain Disks through Garry Clark.

We have recently received a large contribution to this library from Kaypro Users and from the Spectravideo Users Group in Montreal. All of this software was included in our list in the last issue - so look carefully for the new and interesting selection. Thanks to both of those organisations for their recent contribution, and also thanks to all our contributors both local and overseas.

```
10 ' *** MACHINE CODE SPRITE MOVER *** BY ELLIOTT YOUNG
20 ' make mid$(z$,5,1)=# of sprites,     mid$(z$,12,1)=step increment,
mid$(z$,2,1)=1 for x-mov, 0 for y-mov, 3 for col.        mid$(z$,1
1,1)=&Hc6 for -> or down, &Hd6 for <- or up.
30 A=RND(-TIME)
40 SCREEN1,2
50 GOSUB 110
60 DEFUSR=256*PEEK(VARPTR(Z$)+2)+PEEK(VARPTR(Z$)+1)
70 A$=STRING$(32,255):FOR I=0 TO 31:SPRITE$(I)=A$:NEXT
80 FOR I=0 TO 31:PUTSPRITEI,(RND(5)*200,I*5),RND(5)*14+2:NEXT
90 A=USR(0)
100 GOTO 90
110 FOR I=0 TO 26:READ B$:Z$=Z$+CHR$(VAL("&H"+B$)):NEXT
120 RETURN
130 DATA 21,01,1b:'  LD HL,1B01
140 DATA 1e,20:'    LD E,20
150 DATA cd,47,37:'  CALL 3747
160 DATA DB,84:'    IN A,84
170 DATA C6,1:'     ADD A,01
180 DATA f5:'       PUSH AF
190 DATA cd,3c,37:'  CALL 373C
200 DATA F1:'       POP AF
210 DATA D3,80:'    OUT (80),A
220 DATA 01,04,00:'  LD BC,0004
230 DATA 09:'       ADD HL,BC
240 DATA 1d:'       DEC E
250 DATA 20,Eb:'    JR NZ,Eb
260 DATA c9:'       RET
```

This program moves up to 32 sprites in any direction in machine code. Each call of the routine moves the sprites one pixel and then jumps back to basic to allow easy use within a BASIC program. The machine code is stored in a string and then executed using VARPTR, meaning that it is not in any danger of going over a BASIC program in memory.

Z$ can be altered as to which way you want the sprites to move, etc. The second character should equal 1 for x-movement, 0 for y-movement, and 3 for color. The fifth character equals the number of Sprites, currently 32. The eleventh character should equal C6 HEX for right or downward movements or D6 HEX for left or upward movements. The twelfth character is the step increment.

The following two listings from Elliot use the Screen map in locations 6144-6909. The screen is divided into three sections vertically, each made up of 255 8x8 pixel blocks, 32 across by 8 down. Normally each block is filled with a number from 1 to 255 sequentially. This means that every 8x8 block on the screen can be different from any other. But if you vpoke zeroes into every location between 6144 and 6909, then the pattern for each 8x8 square is read from the top left square of each third of the screen. If this top left square is changed then the entire screen changes automatically and instantly. The only problem is a slight flicker that occurs.

In the first listing Elliot uses the first two blocks of each third and alternates them. The border of the screen is appropriately vpoked at the start to change. The idea has great possibilities so do some experimenting!

```
10 REM   USES LOCATIONS 6144-6909 FOR GENERATING A ROTATING B
ORDER FOR                       USE IN TITLE SCREENS,ETC
20 SCREEN1
30 C=15
40 VPOKE 6145,50
50 VPOKE 6145+256,50
60 VPOKE 6145+2*256,50
70 FOR I=1 TO 32 STEP 2:VPOKE 6144+I,0:VPOKE 6144+I+1,1:VPOKE 736+
6144+I,0:VPOKE 736+6144+I+1,1:NEXT
90 FOR I=1 TO 24 STEP2:VPOKE 6144+(I*32),0:VPOKE 6144+(I*32)+32,1:V
POKE 31+6144+(I*32),0:VPOKE 31+6144+(I*32)+32,1:NEXT
110 LINE(0,0)-(7,7),C,BF
120 LINE(0,64)-(7,71),C,BF
130 LINE(0,128)-(7,135),C,BF
140 IF C=15 THEN C=4 ELSE C=15
150 LINE(8,0)-(15,7),C,BF
160 LINE(8,64)-(15,71),C,BF
170 LINE(8,128)-(15,135),C,BF
180 FOR I=1 TO 30:NEXT
190 GOTO 110
```

The second listing uses the same locations and a machine code routine to allow you to draw a picture on the screen while the screen is blank. Then, when the routine is called 3 times (1 for each segment of the screen), the picture appears instantly, in colour. This gives a machine-code like presentation of a title screen, etc.

```
10 ' PROGRAM TO MAKE ENTIRE SCREEN APPEAR AT ONCE, USING LOCATIONS 6144-6
909     AND A MACHINE CODE ROUTINE FOR SPEED.                      PRO
GRAM BY ELLIOTT YOUNG.
20 DEFINT A-Z
30 R=RND(-TIME)
40 SCREEN1
50 GOSUB 140
60 DEFUSR= 256*PEEK(VARPTR(Z$)+2)+PEEK(VARPTR(Z$)+1)
70 FOR I=6144 TO 6909:VPOKE I,0:NEXT'SET UP SCREEN
80 BEEP
90 FOR I=1 TO 50:LINE(RND(5)*245+10,RND(5)*191)-(RND(5)*245+10,RND(5)*191),RN
D(5)*14+2:NEXT'DRAW LINES INVISIBLY
100 BEEP
110 OUT &H81,0:OUT &H81,&H18 OR&H40'SET ADRESS IN VRAM
120 A=USR(0):A=USR(0):A=USR(0)
130 GOTO 130
140 FOR I=1 TO 8:READ A$:Z$=Z$+CHR$(VAL("&H"+A$)):NEXT
150 RETURN
160 DATA 3E,0  :'LD A,0
170 DATA D3,80 :'OUT &H80,A
180 DATA 3C    :'INC A
190 DATA 20,FB :'JR NZ,FB
200 DATA C9    :'RET
```

B